Anish Kumar

Leipzig, Germany | Anish Kumar | Google Scholar | ORCID: 0009-0002-2279-4932 |
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Education

July 2023 – April 2026 **PhD in Meteorology**
Institute for Meteorology, Leipzig University, Germany

July 2019 – July 2021 **MSc in Atmospheric Science**
Department of Atmospheric Science, Central University of Rajasthan, India

Employment History

Doctoral Researcher, Leipzig University, Germany Jul 2023 – May 2026

- Investigated the variability, morphology, persistence, and extreme states of the Arctic stratospheric polar vortex using ERA5 reanalysis, observational datasets, ICON simulations, CMIP data, and climate model output, developing strong expertise in atmospheric dynamics, climate variability, predictability, large scale data analysis, and process based interpretation of Earth system behaviour.
- Conducted targeted numerical experiments with the ICON model to examine how orographic and non orographic gravity wave forcing influences stratospheric circulation, vortex strength, ozone variability, and the dynamical response of the atmosphere to small scale forcing, strengthening experience in physically grounded numerical modelling and diagnostic analysis relevant to weather and climate applications.
- Analysed stratospheric aerosol intervention scenarios to assess how external forcing modifies the morphology, strength, and dynamical behaviour of the stratospheric polar vortex, contributing to understanding of climate intervention impacts and interconnected Earth system responses.
- Developed explainable machine learning frameworks for Arctic ozone prediction by combining statistical learning, regression based methods, ensemble prediction, uncertainty assessment, SHAP based interpretation, and dynamical diagnostics, with emphasis on connecting machine learning results to atmospheric processes, physical mechanisms, and climate predictability.

Project Associate I, Central University of Rajasthan, India Sep 2021 – Jun 2023

- Conducted high resolution numerical simulations of severe convective storms, lightning, heavy rainfall, hailstorm events, and related weather hazards using the WRF model.
- Designed and performed systematic sensitivity experiments across physical parameterizations, model resolutions, time steps, and forecast configurations to evaluate model generalization, quantify uncertainty.
- Evaluated WRF simulations against observational datasets, reanalysis products, lightning diagnostics, rainfall observations, and instability indices.
- Applied WRFDA and 3DVAR data assimilation techniques to incorporate observational information into numerical weather prediction experiments, improving the representation of convective systems and strengthening experience with model initialisation, validation, predictability, forecast evaluation, and robust transfer across different weather cases.
- Developed and implemented a real time WRF based forecasting system using Python, Shell scripting, Linux workflows, and automated visualisation tools, integrating model execution, large data processing, figure generation, forecast production, and operational style workflow management for practical environmental prediction applications.

Teaching & Mentoring Experience

Co-supervised MSc, BSc Thesis and Seminar Projects

– **BSc Thesis:** Understanding the relationship between the Arctic Stratospheric Polar Vortex and springtime ozone loss, at Institute for Meteorology, Leipzig University, Germany
– **MSc Thesis:** Sensitivity experiments on the physical configuration of the WRF model for simulating lightning events over Germany, at Institute for Meteorology, Leipzig University, Germany
– **Seminar Project:** Tropospheric response to strong and stable Arctic Stratospheric Polar Vortex events, at Institute for Meteorology, Leipzig University, Germany

Teaching Assistant

– Master's course *"Dynamics of the Middle Atmosphere"*, Institute for Meteorology, Leipzig University, Germany
– Master's course *"Weather Analysis and Visualization Laboratory"*, Department of Atmospheric Science, Central University of Rajasthan, India

Skills

Programming & Scripting	Python (NumPy, xarray, Pandas, Cartopy, matplotlib, scikit-learn), Fortran, Shell Scripting
Numerical Weather & Climate Modelling	ICON, WRF, WRFDA (Data Assimilation), ensemble design, sensitivity studies, stratospheric vortex diagnostics
Machine Learning	Explainable ML (SHapley Additive exPlanations), ensemble prediction, regression models
High-Performance Computing	Slurm job scheduling, MPI, Levante-DKRZ clusters
Professional Skills	Time Management, Problem Solving, Independent Research, Eager to Learn

Publications

Anish Kumar, (2026). "Advances in Understanding of Exceptionally Strong Arctic Stratospheric Polar Vortex Dynamics and Associated Ozone Loss". PhD thesis. Leipzig University. URL: https://nbn-resolving.org/urn:nbn:de:bsz:15-qucosa2-1042663.

Anish Kumar, Khalil Karami, and Christoph Jacobi, (2025). "A 45-Year Climatological Study of Arctic Stratospheric Polar Vortex Dynamics and Morphology using ERA5 Data (1979-2023)". In: *Climate Dynamics* 63.403. DOI: https://doi.org/10.1007/s00382-025-07889-0.

Anish Kumar, Khalil Karami, Christoph Jacobi, and Sina Mehrdad, (2025). "Exploring the impact of orographic and non-orographic gravity waves on arctic stratospheric polar vortex dynamics and springtime ozone loss". In: *Journal of Atmospheric and Solar-Terrestrial Physics* 273, p. 106538. ISSN: 1364-6826. DOI: http://dx.doi.org/10.1016/j.jastp.2025.106538.

Anish Kumar, Joyjit Mandal, Sina Mehrdad, and Christoph Jacobi, (2025). "A Novel Approach to Predict the Arctic Stratospheric Ozone from Stratospheric Polar Vortex Dynamics using Explainable Machine Learnings". In: *Scientific Reports* 15.36473. DOI: https://doi.org/10.1038/s41598-025-24379-9.

Khalil Karami, Christoph Jacobi, and **Anish Kumar**, (2025). "The Morphology of the Stratospheric Polar Vortex Under Stratospheric Aerosol Intervention Scenarios". In: *International Journal of Climatology* 45.8, e8838. DOI: https://doi.org/10.1002/joc.8838.

Vinisha, S. K. Panda, **Anish Kumar**, Unashish Mondal, Gitesh Wasson, and Devesh Sharma, (Jan. 2025). "Improving severe thunderstorm simulations over Bihar and West Bengal, India through assimilation of upper air observations using the 3DVAR of WRF model". In: *Natural Hazards* 121.1, pp. 577–600. DOI: 10.1007/s11069-024-06852-2.

Unashish Mondal, **Anish Kumar**, S. K. Panda, S. Shreelekshmi, Bijit Kumar Banerjee, Devesh Sharma, and Someshwar Das, (Jan. 2025). "Unveiling India's lightning patterns through satellite-based climatology and numerical modelling of severe weather events: A review". In: *Journal of Earth System Science* 134.1, p. 30. DOI: 10.1007/s12040-024-02491-0.

Bhavyasree, S. K. Panda, Gitesh Wasson, Unashish Mondal, **Anish Kumar**, and Devesh Sharma, (Oct. 2024). "Assessing the performance of WRF model in simulating severe hailstorm events over Assam and Bihar, India". In: *Modeling Earth Systems and Environment* 10.5, pp. 6623–6646. DOI: 10.1007/s40808-024-02114-z.

Unashish Mondal, **Anish Kumar**, S. K. Panda, Devesh Sharma, and Someshwar Das, (Mar. 2024). "Comprehensive study of thunderstorm indices threshold favorable for thunderstorms during monsoon season using WRF–ARW model and ERA5 over India". In: *Geoenvironmental Disasters* 11.1, p. 5. DOI: 10.1186/s40677-023-00262-5.

Anish Kumar, Unashish Mondal, Subrat Kumar Panda, and Sakia Shabnam Kader, (Jan. 2024). "Sensitivity of horizontal grid resolution on the simulation of thunderstorms over Bangladesh using Weather Research Forecasting Model". In: *Arabian Journal of Geosciences* 17.1, p. 21. DOI: 10.1007/s12517-023-11820-4.

Aditya Sharma, Devesh Sharma, S. K. Panda, and **Anish Kumar**, (Mar. 2024). "Sensitivity analysis of different parameterization schemes of the Weather Research and Forecasting (WRF) model to simulate heavy rainfall events over the Mahi River Basin, India". In: *Agricultural and Forest Meteorology* 346, p. 109885. DOI: 10.1016/j.agrformet.2023.109885.

Unashish Mondal, S. K. Panda, Bijit Kumar Banerjee, **Anish Kumar**, and Devesh Sharma, (Dec. 2023). "Performance evaluation of lightning potential index and flash count using WRF microphysical parameters over Rajasthan and West Bengal, India". In: *Dynamics of Atmospheres and Oceans* 104, p. 101404. DOI: 10.1016/j.dynatmoce.2023.101404.

Anish Kumar, S. K. Panda, Unashish Mondal, Devesh Sharma, and Someshwar Das, (Nov. 2023). "Assessment of model time step and domain resolution of WRF model to simulate the lightning over Bihar and Rajasthan, India". In: *Modeling Earth Systems and Environment* 9.4, pp. 3959–3984. DOI: 10.1007/s40808-023-01724-3.

Unashish Mondal, S. Sreelekshmi, S. K. Panda, **Anish Kumar**, Someshwar Das, and Devesh Sharma, (Nov. 2023). "Diurnal variations in lightning over India and three lightning hotspots: A climatological study". In: *Journal of Atmospheric and Solar-Terrestrial Physics* 252, p. 106149. DOI: 10.1016/j.jastp.2023.106149.

Anish Kumar, Someshwar Das, and S. K. Panda, (Dec. 2022). "Numerical simulation of a widespread lightning event over north India using an ensemble of WRF modeling configurations". In: *Journal of Atmospheric and Solar-Terrestrial Physics* 241, p. 105984. DOI: 10.1016/j.jastp.2022.105984.

Conferences & Workshops

2025 **American Geophysical Union (AGU)**, New Orleans, Louisiana, USA (15–19 December 2025). Presented: *"A Novel Algorithm to Predict the Ozone from Stratospheric Polar Vortex Dynamics using Explainable Machine Learning"*.

2024 **ASIA OCEANIA GEOSCIENCES SOCIETY (AOGS)**, Pyeongchang, South Korea (23–28 June 2024). Presented: *"Intensity and Duration of the Arctic Stratospheric Polar Vortex (1979–2023) and Comparison with ICON Model Simulations."*.

2024 **ESiWACE3–WarmWorld Summer School on HPC for Climate and Weather Applications**, Helsinki, Finland (27 Aug – 2 Sep 2024). Attended.

2022 **National Training Workshop on Fundamentals of Data Assimilation (NTDA)**, Indian Institute of Tropical Meteorology (IITM), Pune, India (09–21 September 2022). Attended.

2022 **Japan Geoscience Union Meeting (JpGU)**, Chiba City, Japan (22 May – 3 June 2022). Presented: *"Simulation of a lightning event using an ensemble of WRF modelling system configurations: Performances of Lightning Potential Index and Lightning Flash Count."*.